Embeddings in Retrieval-Augmented Generation (RAG)

Embeddings are a fundamental component of Retrieval-Augmented Generation (RAG) systems. An embedding is a numerical vector representation of text that captures its semantic meaning. In a RAG pipeline, documents are divided into smaller chunks, and each chunk is converted into an embedding using a machine learning model. These embeddings are stored in a vector database, enabling efficient similarity searches. When a user submits a query, the query is also transformed into an embedding. The system then compares the query embedding with stored document embeddings to identify the most relevant information. The retrieved content is provided as context to a large language model, which generates a response based on both the query and the retrieved knowledge. This approach helps reduce hallucinations and allows the model to access information beyond its training data. Embeddings enable semantic search, allowing documents to be retrieved based on meaning rather than exact keywords, thereby improving retrieval accuracy and response quality.